N. Ege Saraç

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ege.sarac@cispa.de egesarac.github.io
Google Scholar profile DBLP profile

Employment and Positions

Oct 2025–now **Postdoctoral Researcher**, *CISPA Helmholtz Center for Information Security*, Saarbrücken
Visiting Researcher, *Technical University of Munich (TUM)*, Munich
Marie Skłodowska-Curie Postdoctoral Fellow at CISPA since Oct 2026
Host: Bernd Finkbeiner

Jul–Oct 2023 **Research Intern**, *Austrian Institute of Technology (AIT)*, Vienna
Host: Dejan Ničković

Education

Sep 2019–Sep 2025 **PhD** in Computer Science, *Institute of Science and Technology Austria (ISTA)*, Klosterneuburg
Thesis: *A Monitoring-Oriented Theory and Classification of Quantitative Specifications*
Supervisor: Thomas A. Henzinger

Sep 2014–Jun 2019 **BSc** in Computer Science (Minor: Mathematics), *Sabancı University*, Istanbul

Fellowships and Awards

2026 Marie Skłodowska-Curie Postdoctoral Fellowship, QHyperSTAR (~€202k), *European Comm.*
2026 Nominated for the ISTA Outstanding PhD Thesis Award, *ISTA*
2019 Sakıp Sabancı Award; ranked 1st among 544 graduating students, *Sabancı University*
2018 Logic Mentoring Workshop Student Travel Grant, *ACM SIGLOG*
2017 Scholarship for Student Researchers, *Austrian Exchange Service (OeAD)*
2014 Merit-based full scholarship for BSc studies, *Sabancı University*

Publications

*: Corresponding author. †: Ordered by contribution, alphabetical otherwise.

Journal Articles

[LMCS'25] **Safety and Liveness of Quantitative Properties and Automata**. Udi Boker, Thomas A. Henzinger, Nicolas Mazzocchi, N. Ege Saraç*. *Logical Methods in Computer Science*, Volume 21, Issue 2, 2025.

[ESWA'21] **Boosting Expensive Synchronizing Heuristics**. N. Ege Saraç†, Ö. Faruk Altun, K. Tolga Atam, Sertaç Karahoda, Kamer Kaya, Hüsnü Yenigün. *Expert Systems with Applications*, Volume 167, 2021.

Conference Proceedings

[POPL'27] **TriCycle: Proving Temporal Hyperproperties, Coinductively**. Arthur Correnson, N. Ege Saraç†, Mishel Carelli, Bernd Finkbeiner. *54th Symp. on Principles of Programming Languages*, 2027.

[CONCUR'26] **Monitoring Discounted Sum Properties**. Filip Cano, Thomas A. Henzinger, Konstantin Kueffner, N. Ege Saraç. *37th Intl. Conf. on Concurrency Theory*, 2026.

[CAV'26] **Extending QuAK with Nested Quantitative Automata**. Thomas A. Henzinger, Nicolas Mazzocchi, N. Ege Saraç*, Harun Yılmaz. *38th Intl. Conf. on Computer Aided Verification*, 2026.

[FM'26] **Quantitative Monitoring of Signal First-Order Logic**. Marek Chalupa, Thomas A. Henzinger, N. Ege Saraç*, Emily Yu. *27th Intl. Symp. on Formal Methods*, 2026.

[CONCUR'25] **Quantitative Language Automata**. Thomas A. Henzinger, Pavol Kebis, Nicolas Mazzocchi, N. Ege Saraç*. *36th Intl. Conf. on Concurrency Theory*, 2025. Selected for LMCS special issue.

[TACAS'25] **Automating the Analysis of Quantitative Automata with QuAK**. Marek Chalupa, Thomas A. Henzinger, Nicolas Mazzocchi, N. Ege Saraç*. *31st Intl. Conf. on Tools and Algorithms for the Construction and Analysis of Systems*, 2025.

[ISoLA'24] **QuAK: Quantitative Automata Kit** (invited). Marek Chalupa, Thomas A. Henzinger, Nicolas Mazzocchi, N. Ege Saraç*. *12th Intl. Symp. on Leveraging Applications of Formal Methods*, 2024.

[RV'24] **Approximate Distributed Monitoring Under Partial Synchrony: Balancing Speed & Accuracy**. Borzoo Bonakdarpour, Anik Momtaz, Dejan Ničković, N. Ege Saraç*. *24th Intl. Conf. on Runtime Verification*, 2024. Selected for STTT special issue.

[CONCUR'24] **Strategic Dominance: A New Preorder for Nondeterministic Processes**. Thomas A. Henzinger, Nicolas Mazzocchi, N. Ege Saraç*. *35th Intl. Conf. on Concurrency Theory*, 2024.

[CONCUR'23] **Safety and Liveness of Quantitative Automata**. Udi Boker, Thomas A. Henzinger, Nicolas Mazzocchi, N. Ege Saraç*. *34th Intl. Conf. on Concurrency Theory*, 2023. Selected for LMCS special issue.

[ICALP'23] **Regular Methods for Operator Precedence Languages**. Thomas A. Henzinger, Pavol Kebis, Nicolas Mazzocchi, N. Ege Saraç. *50th Intl. Coll. on Automata, Languages and Programming*, 2023.

[FoSSaCS'23] **Quantitative Safety and Liveness**. Thomas A. Henzinger, Nicolas Mazzocchi, N. Ege Saraç*. *26th Intl. Conf. on Foundations of Software Science and Computation Structures*, 2023.

[RV'22] **Abstract Monitors for Quantitative Specifications**. Thomas A. Henzinger, Nicolas Mazzocchi, N. Ege Saraç*. *22nd Intl. Conf. on Runtime Verification*, 2022.

[LICS'21] **Quantitative and Approximate Monitoring**. Thomas A. Henzinger, N. Ege Saraç*. *36th Ann. ACM/IEEE Symp. on Logic in Computer Science*, 2021.

[RV'20] **Monitorability Under Assumptions** (invited). Thomas A. Henzinger, N. Ege Saraç*. *20th Intl. Conf. on Runtime Verification*, 2020.

[LICS'18] **A Theory of Register Monitors**. Thomas Ferrère, Thomas A. Henzinger, N. Ege Saraç. *33rd Ann. ACM/IEEE Symp. on Logic in Computer Science*, 2018.

Teaching Experience

TUM **The Hitchhiker's Guide to Hyperproperties** (Designer and Instructor): Winter 2026/27
Functional Programming & Verification (Head Teaching Assistant): Summer 2026

ISTA **Formalisms Every Computer Scientist Should Know** (Teaching Assistant): Fall 2023
Foundations of Model Checking (Guest Lecturer): Spring 2023
Formal Methods: Algorithmic Approaches (Teaching Assistant): Spring 2022

Sabancı Univ. **Algorithms** (Teaching Assistant): Fall 2018–Spring 2019
Mathematics & Natural Sciences (Workshop & Peer Study Mod.): Spring 2015–Spring 2018

Research Supervision

2026 **Guangyu Zhu**, MSc thesis (co-supervised with E. Yu), *Leiden University*:
Signal First-Order Logic Neural Control

2025 **Harun Yılmaz**, internship (co-supervised with T. A. Henzinger), *ISTA*:
Extending QuAK with Nested Quantitative Automata [CAV'26]

2024 **Pavol Kebis**, rotation project (co-supervised with T. A. Henzinger), *ISTA*:
Quantitative Language Automata [CONCUR'25]

2022 **Pavol Kebis**, internship (co-supervised with T. A. Henzinger and N. Mazzocchi), *ISTA*:
Regular Methods for Operator Precedence Languages [ICALP'23]

Mentoring and Outreach

2025 **Alumni Speaker**, Graduate School Welcome Event, *ISTA*

2023, 2020 **Student Mentor** for three incoming PhD students, *ISTA*

Open-Source Software

2024–now **QuAK** and **Nested-QuAK**: C++ library and CLI tool for quantitative automata analysis

2025–2026 **QSFO**: Python tool for quantitative monitoring of signal first-order logic

2023–2026 **ApxDistMon**: Approximate monitoring algorithms for distributed systems

2019 **SyncBoost**: GPU-accelerated synchronizing heuristics for finite automata

Academic Service and Leadership

Organizer HYPER'27 Workshop (co-located with CAV'27)

HYPER'26 Workshop (co-located with CAV and CSF at FLoC'26)

Program committee CONCUR'26; CAV'26 (AEC), VMCAI'26 (AEC), CAV'25 (AEC), CAV'24 (AEC)

Reviewer *Conferences:* LPAR'26, CONCUR'25, RV'24 (x2), MFCS'24, FSTTCS'23, ATVA'23, CONCUR'23

Journals: Fuzzy Sets and Systems, Science of Computer Programming, Innovations in Systems and Software Engineering

Books: Principles of Systems Design (Thomas A. Henzinger Festschrift)

PhD examining board at Saarland University for Julian Siber (2026), Raven Beutner (2026), Niklas Metzger (2026), Sebastian Schirmer (2026)

Steering committee Helmholtz Association Postdoc Network (2026–now)

Postdoc representative at CISPA (2026–now)

Admissions reviewer for PhD applications at ISTA (2024, 2023)

Maintainer of the publication database for the T. A. Henzinger Group at ISTA (2020–2025)

Professional Development

2026–2027 COMPASS Mentoring Program, *Helmholtz Association*

2025 ProLehre Onboarding – Effective Teaching, *TUM*

2022 Basics in Didactics – Teaching and Learning in Higher Education, *ISTA*

Invited Research Meetings

2027 *Dagstuhl Seminar* "Towards a Unified Theory and Application of Hyperproperties"

2026 *Autobóz* research camp on logic, automata, and games

Research Talks

Extending QuAK with Nested Quantitative Automata

Jul 2026 38th Intl. Conf. on Computer Aided Verification (CAV)

Quantitative Monitoring of Signal First-Order Logic

Jun 2026 AIT Center for Digital Safety & Security Seminar Series

May 2026 27th Intl. Symp. on Formal Methods (FM)

Quantitative Language Automata

Oct 2025 19th Intl. Conf. on Reachability Problems (RP)

Aug 2025 36th Intl. Conf. on Concurrency Theory (CONCUR)

Approximate Distributed Monitoring under Partial Synchrony

Oct 2024 24th Intl. Conf. on Runtime Verification (RV)

Oct 2024 AIT Center for Digital Safety & Security Seminar Series

Safety and Liveness of Quantitative Properties and Automata

Mar 2025 RWTH Aachen Software Modeling and Verification Group Seminar Series

Feb 2025 CISPA Reactive Systems Group Seminar Series

Jan 2025 TU Dresden Algebraic and Logical Foundations of Computer Science Seminar Series

Sep 2024 18th Intl. Conf. on Reachability Problems (RP)

Sep 2024 TU Wien CPS Research Unit Seminar Series

Sep 2024 16th Alpine Verification Meeting (AVM)

Jun 2024 DEVINE Research Team (Inria and IRISA) Formal Methods Seminar

Oct 2023 AIT Center for Digital Safety & Security Seminar Series

Apr 2023 26th Intl. Conf. on Foundations of Software Science and Computation Structures (FoSSaCS)

Advancing the Theory of Quantitative and Approximate Monitoring

Sep 2022 FBK Embedded Systems Seminar Series

Apr 2022 IMT Lucca It-Matters Seminar Series

Quantitative and Approximate Monitoring

Jun 2021 36th Ann. ACM/IEEE Symp. on Logic in Computer Science (LICS)

Jun 2021 ISTA & TU Wien FORSYTE Joint Seminar Series

Monitorability Under Assumptions

Dec 2020 ISTA & TU Wien FORSYTE Joint Seminar Series